WTW 158 Worksheet 9 on L'Hospital's rule, curve sketching and antiderivatives

A. TEXTBOOK PROBLEMS
p 311 no 9, 15, 17, 25, 31, 53, 55, 57, 59
p 300 no 25, 29
p 321 no 7
p 355 no 1- 21 (odd numbers) 33, 39, 49

B. MORE PROBLEMS
For the function $f(x) = x - 2\sin x$, $0 \le x \le 2\pi$, find:
(1) the intervals on which f is increasing / decreasing
(2) extremes
(3) the intervals on which f is concave up / concave down
(4) possible point(s) of inflection

C. TEST QUESTIONS
1. Find the interval(s) on which the function $f(x) = xe^{-x^2}$ is increasing.

2. $\lim_{x \to 0} (1 + kx)^{\frac{1}{x}} =$

[a] 1	[b] 0	[c] e^k	[d] e^{-k}	[e] doesn't exist

D. SELECTED ANSWERS
A. p 311 9. 6 15. 2 17. $\frac{1}{4}$ 25.3 31. 1 53 $\frac{1}{2}$ 55. ∞ 57. 1 59. e^{-2}
p 300 25 & 29 See textbook pA82
p 321 7. See textbook pA84
p 355 1 - 21 (Odd) See textbook p A90

B.(1) Decreasing on $(0, \frac{\pi}{3})$ and on $(\frac{5\pi}{3}, 2\pi)$, increasing on $(\frac{\pi}{3}, \frac{5\pi}{3})$
(2) $f(\frac{\pi}{3}) = \frac{\pi}{3} - \sqrt{3}$ is a local minimum, $f(\frac{5\pi}{3}) = \frac{5\pi}{3} + \sqrt{3}$ is a local / abs maximum
(3) f is concave up on $(0, \pi)$, f is concave down on $(\pi, 2\pi)$
(4) Inflection point (π, π)

C. 1. $(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ 2. [c]

WTW 158 Worksheet 9

A Textbook Problems

p 311 no 9, 15, 17, 25, 31, 53 55, 57 59

9. $\lim\limits_{x\to 4} \dfrac{x^2-2x-8}{x-4}$

$= \lim\limits_{x\to 4} \dfrac{\cancel{(x-4)}(x+2)}{\cancel{(x-4)}} = 6$

15. $\lim\limits_{t\to 0} \dfrac{e^{2t}-1}{\sin t}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{t\to 0} \dfrac{2e^{2t}}{\cos t} = 2$

17. $\lim\limits_{\theta\to \pi/2} \dfrac{1-\sin\theta}{1+\cos 2\theta}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{\theta\to \pi/2} \dfrac{-\cos\theta}{-2\sin 2\theta}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{\theta\to \pi/2} \dfrac{\sin\theta}{-4\cos 2\theta} = \dfrac{1}{(-4)(-1)} = \dfrac{1}{4}$

25. $\lim\limits_{x\to 0} \dfrac{\sqrt{1+2x}-\sqrt{1-4x}}{x}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{x\to 0} \left(\dfrac{\frac{1}{2\sqrt{1+2x}} \times 2 - \frac{1}{2\sqrt{1-4x}} \times (-4)}{1} \right)$

$= \lim\limits_{x\to 0} \dfrac{1}{\sqrt{1+2x}} + \dfrac{2}{\sqrt{1-4x}} = 3$

31. $\lim\limits_{x\to 0} \dfrac{\arcsin x}{x}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{x\to 0} \dfrac{\frac{1}{\sqrt{1-x^2}}}{1} = \dfrac{1}{1} = 1$

53. $\lim\limits_{x\to 0^+} \left(\dfrac{1}{x} - \dfrac{1}{e^x-1}\right)$ (Form: $\infty-\infty$)

$= \lim\limits_{x\to 0^+} \dfrac{(e^x-1)-x}{x(e^x-1)}$ (Form $\frac{0}{0}$)

$\overset{H}{=} \lim\limits_{x\to 0^+} \dfrac{e^x-1}{e^x+xe^x-1}$ (Form $\frac{0}{0}$)

$= \lim\limits_{x\to 0^+} \dfrac{e^x}{e^x+e^x+xe^x} = \dfrac{1}{2}$

55. $\lim\limits_{x\to \infty} (x-\ln x)$ (Form: $\infty-\infty$)

$= \lim\limits_{x\to \infty} x\left(1-\dfrac{\ln x}{x}\right)$

Then $\lim\limits_{x\to\infty} \dfrac{\ln x}{x}$ (Form $\frac{\infty}{\infty}$)

$\overset{H}{=} \lim\limits_{x\to\infty} \dfrac{\frac{1}{x}}{1} = 0$

So $\lim\limits_{x\to\infty} x\left(1-\dfrac{\ln x}{x}\right) = \infty$

57. $y = x^{\sqrt{x}}$

Let $\ln y = \ln x^{\sqrt{x}} = \sqrt{x}\ln x$

$\therefore \lim\limits_{x\to 0^+} \ln y = \lim\limits_{x\to 0^+} \sqrt{x}\ln x$ (form: $0\times -\infty$)

$= \lim\limits_{x\to 0^+} \dfrac{\ln x}{x^{-1/2}}$ (form $\frac{-\infty}{\infty}$)

$\overset{H}{=} \lim\limits_{x\to 0^+} \dfrac{\frac{1}{x}}{-\frac{1}{2}x^{-3/2}}$

$= -2\lim\limits_{x\to 0^+} \sqrt{x} = 0$

Then $\lim\limits_{x\to 0^+} y = e^0 = 1$

59. $(1-2x)^{1/x}$ (Form $(1)^\infty$)

let $y = (1-2x)^{1/x}$

$\therefore \lim\limits_{x\to 0} \ln y = \lim\limits_{x\to 0} \dfrac{1}{x}\ln(1-2x)$ (Form: $\infty\times 0$)

$= \lim\limits_{x\to 0} \dfrac{\ln(1-2x)}{x}$ (Form $\frac{0}{0}$)

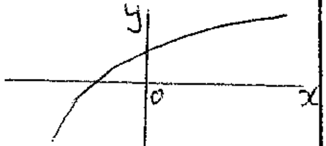
$\overset{H}{=} \lim\limits_{x\to 0} \dfrac{\frac{1}{1-2x}\times -2}{1} = -2$

So $\lim\limits_{x\to 0} y = e^{-2}$

p 300 no 25, 29

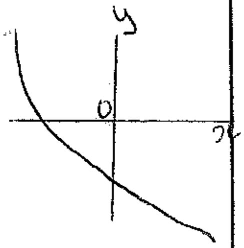
25(a) $f'(x) > 0$ and $f''(x) < 0$ for all x.

- The function f is always increasing ($f' > 0$) and
- Concave downward ($f'' < 0$)



25(b) $f' < 0$ and $f'' > 0$ for all x.

- f is always decreasing ($f' < 0$)
- f is concave up ($f'' > 0$)



29

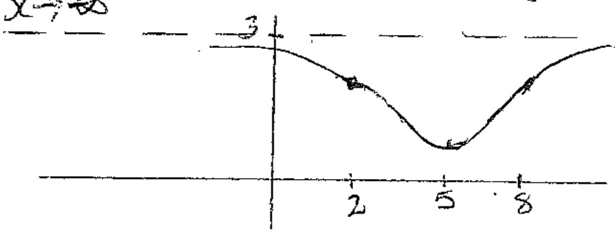
- $f'(5) = 0 \Rightarrow$ horizontal tangent at $x = 5$
- $f'(x) < 0$ when $x < 5$
  $\Rightarrow$ f is decreasing on $(-\infty, 5)$
- $f'(x) > 0$ when $x > 5$
  $\Rightarrow$ f is increasing on $(5, \infty)$
- $f''(2) = 0$, $f''(8) = 0$

f'': < 0 | 2 | > 0 | 8 | < 0

$\therefore$ f is concave upward on $(2, 8)$

f is concave downward on $(-\infty, 2)$ and on $(8, \infty)$

- Inflection points: $(2, f(2))$ and $(8, f(8))$
- $\lim_{x \to \pm\infty} f(x) = 3 \Rightarrow$ HA is $y = 3$

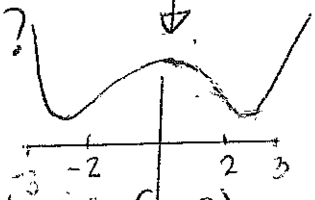


p 321 no 7

$f(x) = \frac{1}{5}x^5 - \frac{8}{3}x^3 + 16x$

1. $D_f = \mathbb{R}$
2. Intercepts
   - $f(0) = 0$
   - $f(x) = 0 \Rightarrow x(\underbrace{\frac{1}{5}x^4 - \frac{8}{3}x^2 + 16}_{= 0})$

   $\Rightarrow x = 0$ or $x = ?$



   only intercept is $(0, 0)$

3. Even / odd

$f(-x) = -\frac{1}{5}x^5 + \frac{8}{3}x^3 - 16x$
$= -\left(\frac{1}{5}x^5 - \frac{8}{3}x^3 + 16x\right)$
$= -f(x)$

f is odd

4. Asymptotes

HA:

$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} x^3\left(\frac{x^2}{5} - \frac{8}{3}\right) + 16x$
$= \infty$

$\lim_{x \to -\infty} f(x) = -\infty$ No HA

VA: $D_f = \mathbb{R}$. No VA

5. Decreasing / Increasing

$f'(x) = x^4 - 8x^2 + 16$
$= (x^2 - 4)(x^2 - 4)$
$= (x-2)^2(x+2)^2 > 0$

f': > 0 | -2 | > 0 | 2 | > 0

f is increasing on $(-\infty, -2)$ and on $(-2, 2)$ and on $(2, \infty)$

6. Extremes

None as f is always increasing

7. Concavity

$f''(x) = 4x^3 - 16x$
$= 4x(x-2)(x+2)$

f'': <0 on $(-\infty,-2)$, >0 on $(-2,0)$, <0 on $(0,2)$, >0 on $(2,\infty)$

f is concave downwards on $(-\infty,-2)$ and on $(0,2)$

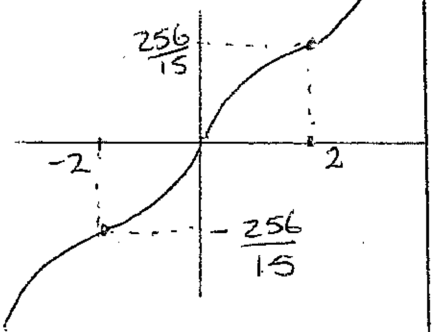
f is concave upwards on $(-2,0)$ and on $(2,\infty)$

8. Inflection Points

$f(-2) = -\frac{256}{15}$, $(0,0)$

and $(2, f(2)) = (2, \frac{256}{15})$

Sketch:



p 355 no 1-21 (odd numbers) 33, 39, 49

1. $f(x) = 4x + 7$
$\Rightarrow F(x) = 2x^2 + 7x + C$

3. $f(x) = 2x^3 - \frac{2}{3}x^2 + 5x$
$\Rightarrow F(x) = \frac{1}{2}x^4 - \frac{2}{9}x^3 + \frac{5}{2}x^2 + C$

5. $f(x) = x(12x+8) = 12x^2 + 8x$
$\Rightarrow F(x) = 4x^3 + 4x^2 + C$

7. $f(x) = 7x^{2/5} + 8x^{-4/5}$
$\Rightarrow F(x) = 5x^{7/5} + 40x^{1/5} + C$

9. $f(x) = \sqrt{2}$
$\Rightarrow F(x) = \sqrt{2}\,x + C$

11. $f(x) = 3\sqrt{x} - 2\sqrt[3]{x}$
$= 3x^{1/2} - 2x^{1/3}$
$\Rightarrow F(x) = 3\left(\frac{2}{3}x^{3/2}\right) - 2\left(\frac{3}{4}x^{4/3}\right) + C$

13. $f(x) = \frac{1}{5} - \frac{2}{x}$
$\Rightarrow F(x) = \frac{1}{5}x - 2\ln|x| + C$

15. $g(t) = \frac{1+t+t^2}{\sqrt{t}} = t^{-1/2} + t^{1/2} + t^{3/2}$
$\Rightarrow G(t) = 2t^{1/2} + \frac{2}{3}t^{3/2} + \frac{2}{5}t^{5/2} + C$

17. $h(\theta) = 2\sin\theta - \sec^2\theta$
$\Rightarrow H(\theta) = -2\cos\theta - \tan\theta + C$

19. $f(x) = 2^x + 4\sinh x$
$\Rightarrow F(x) = \frac{2^x}{\ln 2} + 4\cosh x + C$

21. $f(x) = \frac{2x^4 + 4x^3 - x}{x^3}$, $x > 0$
$= 2x + 4 - x^{-2}$
$\Rightarrow F(x) = \frac{2x^2}{2} + 4x + \frac{1}{x} + C$ $(x > 0)$

33. $f'(t) = \frac{4}{1+t^2}$
$\Rightarrow f(t) = 4\arctan t + C$
$f(1) = 4 \times \frac{\pi}{4} + C = 0$
$\Rightarrow C = -\pi$
so $f(t) = 4\arctan t - \pi$

39. $f''(x) = -2 + 12x - 12x^2$
$\Rightarrow f'(x) = -2x + 6x^2 - 4x^3 + C$
$f'(0) = C = 12$
so $f'(x) = -2x + 6x^2 - 4x^3 + 12$
$f(x) = -x^2 + 2x^3 - x^4 + 12x + D$
$f(0) = 4 = D$
$\therefore f(x) = -x^2 + 2x^3 - x^4 + 12x + 4$

49. $f'(x) = 3 - 4x$

$\Rightarrow f(x) = 3x - 4x^2 + C$

Point : $(2,5)$

So $f(2) = 5 = -2 + C$

$\Rightarrow C = 7$

So $f(x) = 3x - 2x^2 + 7$

Hence $f(1) = 3 - 2 + 7 = 8$

B. More Problems

$f(x) = x - 2\sin x, \ x \in [0, 2\pi]$

1) Decreasing / Increasing

$f'(x) = 1 - 2\cos x = 0$

$\cos x = \frac{1}{2}$

$x = \pi/3, \ x = 5\pi/3$

f': <0 on $(0, \pi/3)$, >0 on $(\pi/3, 5\pi/3)$, <0 on $(5\pi/3, 2\pi)$

f is decreasing on $(0, \pi/3)$ and on $(5\pi/3, 2\pi)$

f is increasing on $(\pi/3, 5\pi/3)$

2) Extremes :

$f(0) = 0 \quad f(2\pi) = 2\pi \simeq 6.28$

$f(\pi/3) = \pi/3 - 2\sin\pi/3 = \pi/3 - \sqrt{3}$ is a local / absolute min

$f(5\pi/3) = 5\pi/3 - 2\left(-\frac{\sqrt{3}}{2}\right)$ is a $= 5\pi/3 + \sqrt{3}$ local / absolute max

3) Concavity

$f''(x) = 2\sin x = 0$

$\Rightarrow \sin x = 0$

$\Rightarrow x = \pi, \ x = 0, 2\pi$

f'': >0 on $(0, \pi)$, <0 on $(\pi, 2\pi)$

f concave upwards on $(0, \pi)$

f concave downwards on $(\pi, 2\pi)$

4. Inflection Point

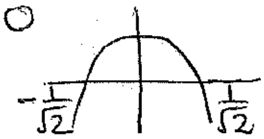
$(\pi, f(\pi)) = (\pi, \pi)$

C. Test Questions

1. $f(x) = xe^{-x^2}$

$f'(x) = \underbrace{e^{-x^2}}_{>0}(1 - 2x^2) = 0$

$\Rightarrow 1 - 2x^2 = 0$



$f'(x) > 0 \Rightarrow x \in \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

So f is increasing on $\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

2. $\lim_{x \to 0} (1 + kx)^{1/x}$ (Form 1^∞)

Let $y = (1 + kx)^{1/x}$

$\Rightarrow \ln y = \frac{1}{x}\ln(1 + kx)$

Then $\lim_{x \to 0} \ln y = \lim_{x \to 0} \frac{\ln(1 + kx)}{x}$ (Form $\frac{0}{0}$)

$\stackrel{H}{=} \lim_{x \to 0} \frac{k}{1 + kx}$

$= \frac{k}{1}$

So $\lim_{x \to 0} y = e^k$

answer : [c]